

Navigation in GNSS-Denied Environments Using MEMS-Grade Sensors and Geophysical Anomalies: A UKF Approach

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Introduction

The GNSS Dependency Problem

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- Indoor/underground environments
- Underwater operations
- Ionospheric disturbances and solar weather

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Intentional Disruption:

- Electronic warfare (jamming)
- GNSS spoofing attacks
- Critical for aviation, maritime, military

Solution Needed

Alternative PNT (Alt-PNT) solutions that do not rely on external RF signals

Geophysical Navigation: The Cost-Performance Gap

Core Concept:

- Exploit naturally occurring variations in Earth's physical fields
- Gravity and magnetic anomalies are spatially correlated
- Maps available for reference
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- Navigation/Tactical-grade IMUs with dedicated gravimeters and magnetometers
- Cost: \$10K - \$100K+
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Potential Impact: Extends alt-PNT geophysical navigation to new domains

- Commercial drones
- Small autonomous vehicles
- Low SWaP constraints

Can we extend geophysical navigation to low-Size Weight and Power (SWaP) mass-market platforms?

Methodology

Data Collection:

- Smartphone-grade MEMS sensors
- Highway driving scenarios
- 30 minutes to several hours each
- Tens to hundreds of kilometers

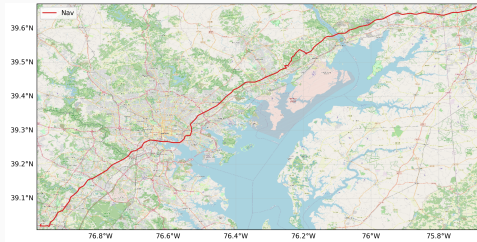
MEMS-Nav Dataset

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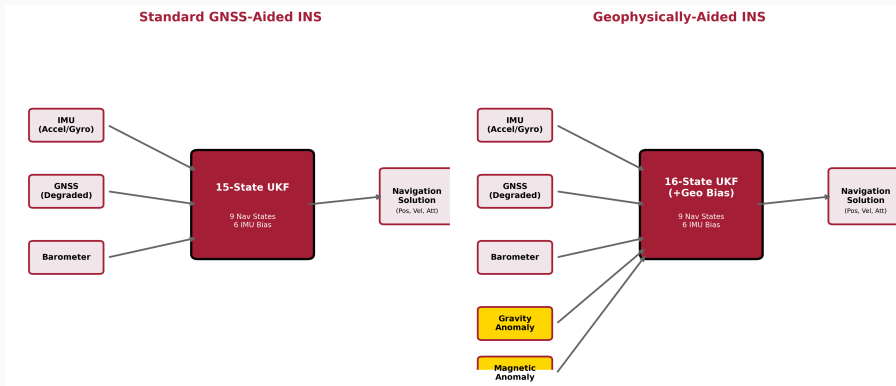
Sensor Suite:

- 3-axis accelerometer (m/s^2)
- 3-axis gyroscope (rad/s)
- 3-axis magnetometer (μT)
- GNSS receiver (lat, lon, alt, velocity)
- Barometric altimeter



*An example trajectory from the
MEMS-Nav dataset*

System Architecture



15-State UKF INS + Geophysical Augmentation

Standard 15-State UKF INS

State Vector (15 states):

$$\mathbf{x} = \begin{bmatrix} L & \lambda & h & v_n & v_e & v_d & \phi & \theta & \psi & b_{a_x} & b_{a_y} & b_{a_z} & b_{g_x} & b_{g_y} & b_{g_z} \end{bmatrix}^T$$

Position (LLF):

- L - Latitude
- λ - Longitude
- h - Altitude (MSL)

Velocity (NED):

- v_n - North
- v_e - East
- v_d - Down

Attitude + Biases:

- ϕ, θ, ψ - Roll, Pitch, Yaw
- $\mathbf{b}_a$ - Accel biases
- $\mathbf{b}_g$ - Gyro biases

Propagation: Strapdown mechanization equations (Groves Ch. 5)

Updates: GNSS position/velocity + barometric altitude

AR(1) Correlated Errors:

- Position: $\rho_{\text{pos}} = 0.99$, $\sigma_{\text{pos}} = 5$ m
- Velocity: $\rho_{\text{vel}} = 0.95$, $\sigma_{\text{vel}} = 5$ m/s
- Slowly drifting biases (not white noise)
- Models multipath/atmospheric delays

GNSS Degradation Scenario

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Covariance Inflation:

- Scale factor: $r_{\text{scale}} = 15$
- 15× advertised accuracy
- Simulates poor DOP reporting
- Filter trusts GNSS less

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Challenge

Combination of correlated errors + inflated covariance stresses filter's ability to maintain accurate state estimates

Applied identically to both baseline and geophysically-aided systems

Anomaly Maps:

- **Gravity:** IGPP Earth Free-air Anomaly, 1 arc-minute resolution (~ 1.8 km)
- **Magnetic:** WDMAM, 3 arc-minute resolution (~ 5.6 km)
- Accessed via GMT software
- Bilinear interpolation

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Processing:

1. Measure gravity/magnetic field
2. Subtract theoretical model
3. Compare residual to anomaly map
4. Add bias state to UKF (16 states)

Generalized Measurement Model:

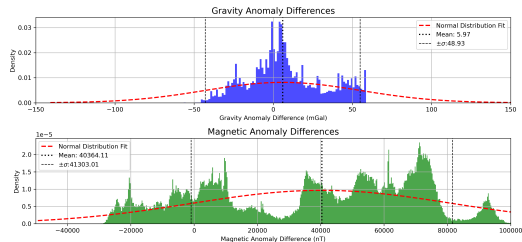
$$h(\mathbf{x}) = \text{map}(L, \lambda) + \mathcal{N}(0, \sigma_{\text{geo}})$$

Table 1: Measurement residual distributions

Anomaly Type	Mean	Std. Dev
Gravity	5.97 mGal	48.93 mGal
Magnetic	40,364 nT	41,303 nT

Measurement Noise Parameters:

Anomaly Differences between Measured and Map Values



Results

Three Configurations Tested:

1. Standard GNSS-aided INS (degraded GNSS baseline)
2. GNSS + Gravity anomaly aiding
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Error Metrics:

- Haversine distance (great-circle distance on sphere)
- Altitude difference
- Compare estimated position to reference GNSS
- $\Delta \text{Error} = \text{Geophysical Error} - \text{Baseline Error}$; negative values indicate improvement

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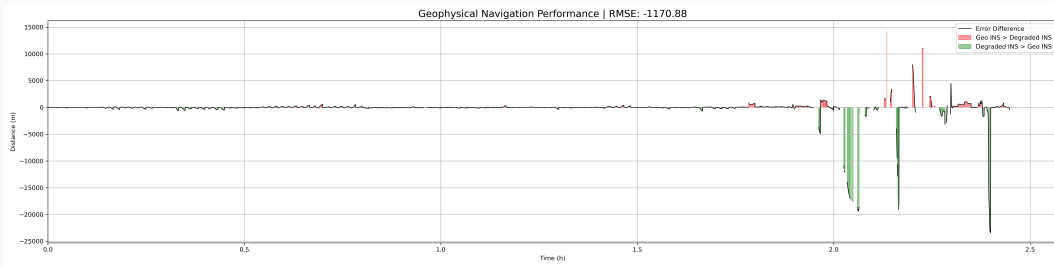
Statistics: RMSE, Mean, Median position error; individual and aggregated across 13 trajectories

Summary Statistics Across All Trajectories

Trajectory Name	Gravity Aiding			Magnetic Aiding		
	RMSE	Mean	Median	RMSE	Mean	Median
2023-08-04_214758	-1170.88	-168.35	-5.98	95.24	-12.69	-11.15
2023-08-06_144805	-72.95	-61.85	-22.26	-72.69	-64.26	-20.25
2023-08-09_163741	8599.73	1133.66	40.57	8342.82	1033.11	-9.30
2025-03-01_150426	154.45	162.26	89.34	86.87	60.86	-9.54
2025-03-01_164639	111.75	129.15	45.96	39.65	31.49	-14.00
2025-06-18_15-09-25	24.24	34.21	19.81	50.27	7.81	-5.17
2025-06-18_16-52-32	58.32	27.63	-0.23	15.04	43.64	-0.42
2025-06-27_11-54-35	-565.99	-318.34	-165.42	-550.91	-402.55	-215.20
2025-07-04_17-24-46	40.81	-87.90	-60.41	19.59	12.08	-17.50
2025-07-18_23-13-43	-8.42	-4.25	-0.97	222.10	100.38	6.28
2025-07-31_23-36-03	-688.07	-472.63	-107.97	-687.96	-471.84	-101.67
2025-09-26_19-03-38	285.61	219.70	34.73	21.73	16.94	9.53
2025-09-27_18-10-16	-359.47	-132.02	-23.49	-10.58	-3.28	-2.00
median	32.53	11.69	-3.47	30.69	14.51	-10.34

Results: Gravity Aiding Success

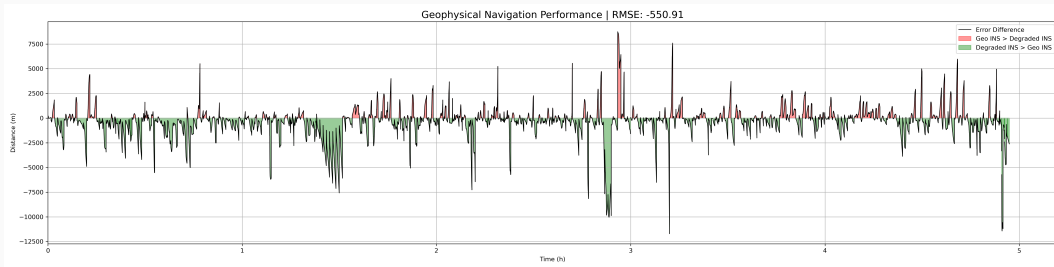
Trajectory 2023-08-04_214758



RMSE Improvement: -1,170 meters

Results: Magnetic Aiding Success

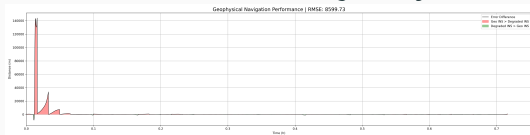
Trajectory 2025-06-27_11-54-35



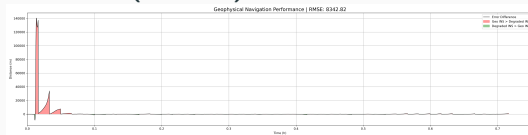
RMSE Improvement: -550 meters

Results: Failure Mode Analysis

Trajectory 2023-08-09_163741 (Outlier)



Gravity: +8,659m error



Magnetic: +8,635m error

- Catastrophic initial divergence followed by recovery
- Indicates need for integrity monitoring and robust initialization
- **Need:** Improved filter robustness and adaptive measurement weighting

Discussion

When Does Geophysical Aiding Work?

Success Factors:

- Strong geophysical gradients
- Well-mapped regions
- Favorable local anomaly characteristics
- Complementary modalities (gravity vs magnetic)

Performance Range:

- Best cases: 100-1,100 m improvement
- Typical: 10-100 m improvement
- Median errors consistently better

Limitations:

- MEMS sensor noise
- Map resolution (1-3 arc-min)
- Weak anomaly fields
- Filter divergence risk

Key Insight

Different trajectories benefit from different modalities - multi-modal fusion could improve consistency

Low-Cost Alt-PNT is Feasible:

- Demonstrated with extremely low-cost MEMS sensors (smartphone-grade)
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Conclusion

First demonstration of geophysical navigation feasibility using only:

- MEMS-grade sensors
- Open-source geophysical maps
- Smartphone-class hardware

Key Findings

First demonstration of geophysical navigation feasibility using only:

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Performance Highlights

- Error reductions: tens to hundreds of meters on favorable trajectories
- Best case: $\approx 1,100\text{m}$ RMSE improvement
- Median improvements: 3 to 10 meters across all trajectories
- Despite MEMS noise and limited map resolution

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Opens pathway for GNSS-denied geophysical navigation from niche high-end military/aerospace applications to low-SWaP military, aerospace, maritime, and commercial applications.

Immediate Priorities:

1. Multi-Modal Fusion

- Combine gravity + magnetic measurements
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4. Observability Analysis

- Characterize which regions/trajectories benefit most
- Map-matching performance metrics

Acknowledgments

Data Collection Support:

- Beth Brodovsky
- Jeremy Brodovsky
- Monica Brodovsky
- Bill Siegl
- Alkesh Kumar Srivastava

Resources:

- MEMS-Nav Dataset: <https://doi.org/10.5281/zenodo.17582434>
- Strapdown-rs Software: <https://github.com/jbrodovsky/strapdown-rs>
- IGPP Gravity Anomaly Maps (EGM2008)
- World Digital Magnetic Anomaly Map (WDMAM)
- Generic Mapping Tools (GMT)

Questions?

James Brodovsky: jbrodovsky@temple.edu

Brodovsky, J., & Dames, P. "Navigation in GNSS-denied environments using MEMS-grade sensors and geophysical anomalies: A UKF approach." *ION International Technical Meeting 2026*.

Backup Slides

Unscented Kalman Filter Steps:

Sigma Point Generation:

$$\mathcal{X}(\mathbf{x}, \mathbf{P}) = \begin{cases} \mathcal{X}_0 = \mathbf{x} \\ \mathcal{X}_i = \mathbf{x} + \left(\sqrt{(n + \lambda) \mathbf{P}} \right)_i, & i = 1, \dots, n \\ \mathcal{X}_{i+n} = \mathbf{x} - \left(\sqrt{(n + \lambda) \mathbf{P}} \right)_i, & i = 1, \dots, n \end{cases}$$

Backup: UKF Algorithm ii

Prediction:

1. Generate sigma points: $\mathcal{X}_{t-1}$
2. Propagate through dynamics: $\bar{\mathcal{X}}_t = g(\mathcal{X}_{t-1}, \mathbf{u}_{t-1})$
3. Compute predicted mean: $\bar{\mathbf{x}}_t = \sum W^{(m)} \bar{\mathcal{X}}_t$
4. Compute predicted covariance: $\bar{\mathbf{P}}_t = \sum W^{(c)} (\bar{\mathcal{X}}_t - \bar{\mathbf{x}}_t)(\bar{\mathcal{X}}_t - \bar{\mathbf{x}}_t)^\top + \mathbf{Q}$

Update:

1. Predict measurements: $\mathcal{Z}_t = h(\mathcal{X}_t)$
2. Innovation: $\mathbf{y}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$
3. Kalman gain: $\mathbf{K}_t = \mathbf{C}_t \mathbf{S}_t^{-1}$
4. State update: $\mathbf{x}_t = \bar{\mathbf{x}}_t + \mathbf{K}_t \mathbf{y}_t$

Backup: Strapdown Mechanization

Attitude Update:

$$\mathbf{C}_{t+1} \approx \mathbf{C}_t (I + \boldsymbol{\Omega}_b \Delta t) - (\boldsymbol{\Omega}_i + \boldsymbol{\Omega}_n) \mathbf{C}_t \Delta t$$

Velocity Update:

$$\mathbf{v}_{t+1} \approx \mathbf{v}_t + \left(\hat{\mathbf{f}}_t + \mathbf{g}_t - (2\boldsymbol{\Omega}_i + \boldsymbol{\Omega}_n) \mathbf{v}_t \right) \Delta t$$

Position Update:

$$h_{t+1} = h_t + \frac{1}{2} (v_{d,t} + v_{d,t+1}) \Delta t$$

$$L_{t+1} = L_t + \frac{1}{2} \left(\frac{v_{n,t}}{R_n + h_t} + \frac{v_{n,t+1}}{R_n + h_{t+1}} \right) \Delta t$$

$$\lambda_{t+1} = \lambda_t + \frac{1}{2} \left(\frac{v_{e,t}}{(R_e + h_t) \cos L_t} + \frac{v_{e,t+1}}{(R_e + h_{t+1}) \cos L_{t+1}} \right) \Delta t$$